

Recursion and Immutable Data

CS 350

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THESE ARE YOUR
FATHER'S PARENTHESSES



ELEGANT
WEAPONS



FOR A MORE... CIVILIZED AGE.

Overview

Objectives

- To learn how to use Dr. Racket

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- To learn how to install and use `#lang flit`

Programming in CS 350

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- The Dr. Racket editor

Racket

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- Language for making languages
 - You will find documentation for many of these languages online
 - They are different from what we are doing. Beware!

What is Dr. Racket?

- IDE for Racket

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 - Feedback when writing code
 - Can evaluate expressions while you're writing your code

Flit

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 - You can do a lot with very little

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- Data Types with Variants

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 - NOT always a named function!

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- Languages like C++ and Python use “infix notation” for math

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- In Flit, you write `( f x y z )`

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- In C++ you write `f(x, y, z)`
- In Flit, you write `(f x y z)`
- 99% of what you do in Flit is calling functions

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- JSON is basically just funny-looking parentheses

Example

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(+ 2 7)
(- 10 0.5)
(* 1/3 2/3)
(/ 1 100000000000000.0)
(max 10 20)
(modulo 10 3)
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9.5
2/9
1e-12
20
1
```

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(> (/ 0 1) 1)
(zero? (- (+ 1 2) (+ 3 0)))
(and (< 1 2) (> 1 0))
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 - *Mutate* value of variable

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- Use functions and recursion to do all computation

Example: Imperative Conditionals

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```
if (someCondition)
  doThis
else
  doThat
```

- The if-statement doesn't have a value, but it can mutate (change) the state

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  (if (= 2 (+ 1 1))  
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 - e.g. Above, the 2nd if evaluates to 3 because the condition evaluates to true

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- Example:

Imperative Breaks Equational Reasoning

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```
int x;  
x = 3;  
x = 4;
```

- Have $x = 3$ and $x = 4$, so surely $3 = 4$, right?

- We can describe the behaviour of if-expressions with the following two equations:

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(define (addOne [x : Number]) : Number  
  (+ x 1))  
(addOne 10)
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11
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Functions

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(define (isRemainder [x : Number]
                    [y : Number]
                    [remainder : Number])
  : Boolean
  (= remainder (modulo x y)))
(isRemainder 10 3 1)
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- General form:

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Functions (ctd.)

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```
(define (functionName  
    [argName : argType]  
    ...  
    [argNameN : argTypeN]) : returnType  
  functionBody)
```

- Later in the course we'll see another way of defining functions

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 - Compares pointers, so very fast

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 - Same, as long as there's no side effects

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- `letrec` : multiple definitions that can all refer to each other
 - We'll see this later when we learn about lambdas

Functional Thinking and Recursion

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 - Mapping from inputs to outputs

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 - But lots aren't

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Further reference:

Factorial - Representation

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```
(define (factorial [n : Number]) : Number  
  (error 'factorial "TODO"))
```

Factorial - Examples

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```
(test (factorial 0) 1 )  
(test (factorial 1) 1 )  
(test (factorial 2) 2 )  
(test (factorial 3) 6 )  
(test (factorial 4) 24 )  
(test (factorial 5) 120 )
```

- Notice the pattern?

Factorial - Template

- A natural number is either

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```
(define (factorial [n : Number]) : Number
  (if (zero? n)
      (error 'zero "TODO")
      (let ([n-1 (- n 1)])
        (error 'suc "TODO")))
  ))
```


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Factorial - Filling holes

- Example gives the base case for 0

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```
(define (factorial [n : Number]) : Number
  (if (zero? n)
      1
      (let*
        ([n-1 (- n 1)]
         [fn-1 (factorial n-1)])
        (* n fn-1)))
  ))
```

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(test (factorial 0) 1 )  
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```
good (factorial 0) at line 11  
  expected: 1  
  given: 1
```

```
good (factorial 5) at line 12  
  expected: 120  
  given: 120
```

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 - Aside: I research languages where you *can* express this with types

Another Example: Exponentiation

- Live coding in Dr. Racket

Unbounded Data: Lists

Functional Linked Lists

- Every linked list is one of:

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- `Cons` does **not** change its input
 - Creates a new list whose tail is the old list

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- Lots more helper functions, see the documentation

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```
(define (list-template
  [xs : (Listof Number)])
  (if (empty? xs)
      (error 'nil "TODO")
      (let ([h (first xs)]
            [t (rest xs)]
            [tRet (list-template t)])
        (error 'cons "TODO")))
  ))
```


Example: Sum

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```
(define (sum [xs : (Listof Number)])  
  : Number  
  (if (empty? xs)  
      0  
      (let* ([h (first xs)]  
              [t (rest xs)]  
              [tRet (sum t)])  
        (+ h tRet))))  
(sum '())  
(sum '(1 2 3))  
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0

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 - e.g. `first '()` will raise an error

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(define (increment [xs : (Listof Number)])  
  : (Listof Number)  
  (type-case (Listof Number) xs  
    [empty  
     empty]  
    [(cons h t)  
     (cons (+ h 1) (increment t))]))  
(increment '(2 3 4))
```

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```
'(3 4 5)
```


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- Output has type 'a

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- Like symbols
 - Plait type inference figures out solutions for type variables when you call a function
 - E.g. `first : ((Listof 'a) -> 'a)`
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- Output has type 'a
- e.g. `first '(1 2 3)` is a Number, but `first '(#t #f #t)` is a Boolean

Parametric Polymorphism

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 - e.g. `first '(1 2 3)` is a Number, but `first '(#t #f #t)` is a Boolean
- Later, this will be very useful for writing generic list operations

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```
(define (concat [xs : (Listof 'elem)]  
            [ys : (Listof 'elem)])  
  : (Listof 'elem)  
  (type-case (Listof 'elem) xs  
    [empty  
     ys]  
    [(cons h t)  
     (cons h (concat t ys))]))
```

Example: List Concatenation (ctd.)

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'(1 2 3 4 5 6)  
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```

More Examples

- Demo: Dr. Racket (as time permits)

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 - Filtering out odd elements of a list